Preeti Hegde

Software Engineer | AI | Microservices

AI Master's graduate and Software Engineer with experience in **machine learning, NLP, and data-driven systems**. I enjoy building *intelligent solutions* that support *real-world decision-making*. With a strong foundation in **backend development and system design**, I focus on creating **scalable, production-ready applications**. I bring a *curious mindset*, strong **problem-solving skills**, and a passion for *practical AI*.

Bengaluru, India

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WORK EXPERIENCE

Data Analyst Intern
Internship | Nov 2025 – Jan 2026 | Remote
Chi2 Data Analytics and Artificial Intelligence GmbH

Data Analysis and Research Workflows Python | Statistical Analysis

- Analyzed structured research datasets using **statistical inference and predictive modelling** to support data-driven scientific decision-making.
- Developed **data preprocessing pipelines** to standardize experimental datasets and improve reproducibility of analysis workflows.
- Documented analytical procedures and results to ensure **transparent and reproducible research processes**.

Master Thesis
Research | Nov 2024 – May 2025 | Würzburg, Germany
Technische Hochschule Würzburg-Schweinfurt

Computer Vision Research Python | PyTorch | Deep Learning

- Designed a **multi-view deep learning pipeline** for classification and anomaly detection using industrial image datasets captured from five camera angles.
- Evaluated early and late fusion strategies; achieved **96% classification accuracy** and **0.85 AUROC** for anomaly detection.
- Containerised the research pipeline using **Docker** to enable reproducible experiments and scalable model evaluation.

Software Engineer
Full-time | Jan 2021 – Mar 2023 | Bengaluru, India
Altisource

Workflow Orchestration BPMN (Camunda 7)

- Designed and implemented end to end workflow orchestration using **Camunda 7**, enabling dynamic task routing, state management, and process automation across distributed systems.
- Modeled and deployed business processes using **BPMN 2.0** with user tasks, service tasks, and event driven workflows for scalable execution.

Backend Systems Java | Spring Boot | SQL | REST APIs

- Designed and maintained **Spring Boot microservices** for distributed backend systems processing high-volume transactional data.
- Developed and integrated **REST and SOAP APIs** enabling real-time data exchange with internal platforms and external vendor systems.
- Implemented **Stripe payment gateway integration using webhooks**, improving payment traceability and reducing failures by **40%**.
- Built structured validation and error-handling mechanisms, reducing recurring production incidents by **25%**.

Cloud and DevOps AWS | Docker | CI/CD

- Containerized and deployed **10+ backend services** using Docker on **AWS ECS and EKS**, improving scalability and runtime reliability.
- Developed and maintained **CI/CD pipelines** using Bitbucket and Bamboo, accelerating release cycles by **30%**.

EDUCATION

Master of Science – Artificial Intelligence *Mar 2023 – Jun 2025*
Technische Hochschule Würzburg-Schweinfurt *Würzburg, Germany*

Bachelor of Engineering – Computer Science *Aug 2016 – Jun 2020*
Sri Siddhartha Institute of Technology *Tumakuru, India*

TECHNICAL SKILLS

Programming
Java, Python

Backend Frameworks
Spring Boot, Spring Framework, Spring MVC, Spring Batch, REST, Camunda 7/8.5, Flowable

API, Messaging & Integration
REST APIs, SOAP APIs, Kafka, JSON, XML

Tools, Methodologies & DevOps
AWS, Docker, CI/CD Pipelines, Git, Maven, Agile

Machine Learning & Frameworks
Deep Learning, Computer Vision, Natural Language Processing, Generative AI, PyTorch, Transformer Architectures

LLM & GenAI
LangChain, LangGraph, LLM API's, ONNX, GPU Acceleration (CUDA), RAG, Prompt Engineering, Vector Databases

Databases
PostgreSQL, MySQL, FAISS, Qdrant

Languages
English, Kannada, Hindi, German

PROJECTS

- Production Grade RAG Platform for Document Intelligence
- SummariseIt: AI Text Summarization Tool
- Reinforcement Learning Blackjack Agent
- Sudoku Solver using Deep Learning

PUBLICATIONS

Feedback and Recommendation System using Natural Language Processing *June 2020*
Authors: Sangeetha D R, **Preeti V Hegde**, Prerana N G, Manjunatha Kumar B H

CERTIFICATES

- LLM Foundations: Vector Databases for Caching and RAG
- Model Context Protocol (MCP) for Beginners by Microsoft
- Core Java Certification